

1 (1) 3

2 (4) no solution

3 (1) 10 units

4 (2) prepositional

5 (2) $-\frac{1}{2}$

6 (1) x^3y^2

7 Given poly. $4x^2 - 2x + (k-4)$
 here $a = 4$
 $b = -2$
 $c = k-4$

Sum of roots = $-\frac{b}{a} = -\left(\frac{-2}{4}\right) = \frac{2}{4} = \frac{1}{2}$

Product = $\frac{c}{a} = \frac{k-4}{4}$

A/C to question

Sum of roots = $\frac{1}{2}$ × product of roots

$$\frac{1}{2} = \frac{1}{2} \times \left(\frac{k-4}{4}\right)$$

$$\frac{1}{2} = \frac{k-4}{8}$$

$$8 = 2k - 8$$

$$8 + 8 = 2k$$

$$16 = 2k$$

$$\boxed{k = 8}$$

Prob 20

$$\frac{dP}{dt} = \frac{P}{10} - \frac{P^2}{1000} \quad (1)$$

$$\text{Let } P = \frac{1000}{u} \Rightarrow \frac{dP}{dt} = -\frac{1000}{u^2} \frac{du}{dt}$$

$$\text{Sub in (1)}$$

$$\frac{1000}{u^2} \frac{du}{dt} = \frac{1000}{u} - \frac{1000^2}{1000 u^2}$$

$$\frac{du}{dt} = u - \frac{1000}{u}$$

$$\frac{du}{u^2 - 1000} = dt$$

$$\int \frac{du}{u^2 - 1000} = \int dt$$

$$\frac{1}{2\sqrt{1000}} \ln \left| \frac{u - \sqrt{1000}}{u + \sqrt{1000}} \right| = t + C$$

$$\ln \left| \frac{u - \sqrt{1000}}{u + \sqrt{1000}} \right| = 2\sqrt{1000}(t + C)$$

$$\frac{u - \sqrt{1000}}{u + \sqrt{1000}} = e^{2\sqrt{1000}(t + C)}$$

$$\frac{u - \sqrt{1000}}{u + \sqrt{1000}} = e^{2\sqrt{1000}t} \cdot e^{2\sqrt{1000}C}$$

$$\frac{u - \sqrt{1000}}{u + \sqrt{1000}} = e^{2\sqrt{1000}t} \cdot K$$

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Equating both sides

$$(53)^2 = (pq)^2$$

$$3 = p^2 q^2$$

$$3q^2 = p^2 \quad \text{--- (1)}$$

$\Rightarrow 3$ divides p^2

$\Rightarrow 3$ divides p --- (2)

Let $p = 3m$ for any true integer from equation (1)

$$3q^2 = (3m)^2$$

$$3q^2 = 9m^2$$

$$q^2 = 3m^2$$

3 divides q^2

3 divides q --- (3)

So from (1) & (3)

3 is common factor of p & q

But this is contradiction to our assumption

So, our assumption is wrong

So 53 is an irrational number

19. Let the fraction be $\frac{x}{y}$

Let sum $\frac{x+y}{y} = 1$

$\frac{x+y}{y} = 1 \Rightarrow x+y = y \Rightarrow x = 0$ (1)

Let sum $\frac{x+y}{y} = \frac{1}{2}$ (2)

$\frac{x+y}{y} = \frac{1}{2} \Rightarrow 2x+2y = y \Rightarrow 2x+y = 0$

Let equation (1) & (2) be

$x = 0$
 $y = 0$

Now solving a value of x in (2)

$2x + 2y = y \Rightarrow 2x = -y$

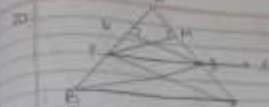
$2x = -y \Rightarrow 2x = -0$

$2x = 0 \Rightarrow x = 0$

Now $\frac{x+y}{y} = 1 \Rightarrow \frac{0+y}{y} = 1$

Let sum of x & y fraction $\frac{x}{y}$

Let fraction $\frac{x}{y} = \frac{1}{2}$



Mathematical: If a line segment is drawn from a vertex of a triangle to a point on the opposite side, then the ratio of the area of the triangle formed by the vertex and the point on the opposite side to the area of the whole triangle is equal to the ratio of the length of the segment on the opposite side to the length of the opposite side.

Given: In $\triangle ABC$, D is a point on BC such that $\frac{BD}{BC} = \frac{1}{2}$

To prove: $\frac{[ABD]}{[ABC]} = \frac{1}{2}$

Proof: Let us draw $DE \parallel AC$ such that E is on AB

In $\triangle ABC$, $DE \parallel AC$ $\therefore \frac{BD}{BC} = \frac{BE}{BA}$ (1)

Also, $\frac{BD}{BC} = \frac{1}{2}$ (Given) $\therefore \frac{BE}{BA} = \frac{1}{2}$ (2)

Now, $\frac{[ABD]}{[ABC]} = \frac{BE}{BA} = \frac{1}{2}$ (3)

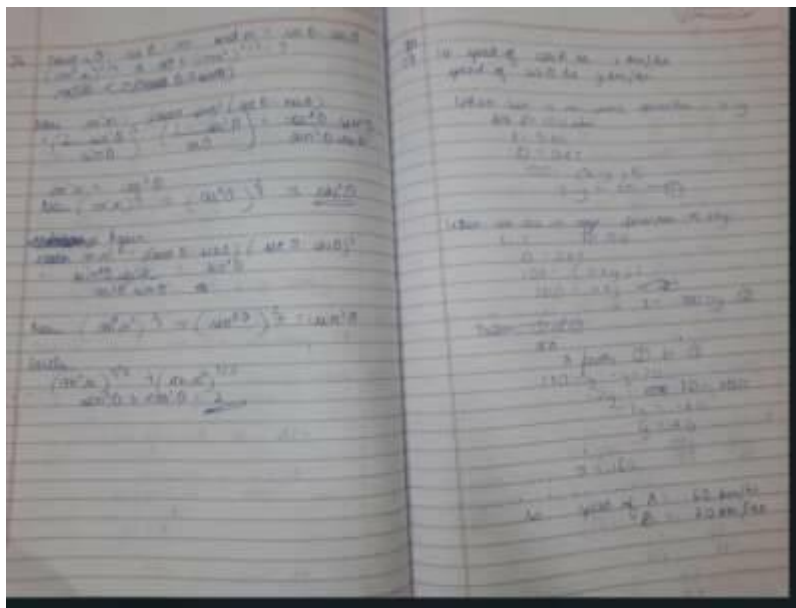
Thus, $\frac{[ABD]}{[ABC]} = \frac{1}{2}$ (Proved)

Let sum $\frac{x}{y} = \frac{1}{2}$

Let sum $\frac{x}{y} = \frac{1}{2}$

1. $\frac{1}{2}$
 2. $\frac{1}{3}$
 3. $\frac{1}{4}$
 4. $\frac{1}{5}$

21. A well known fact is that $\frac{1}{x^2} = x^{-2}$.
 $\frac{d}{dx} x^{-2} = -2x^{-3} = -\frac{2}{x^3}$
 $\frac{d}{dx} \frac{1}{x^2} = -\frac{2}{x^3}$
 Verify by using the quotient rule.
 ~~$\frac{d}{dx} \frac{1}{x^2} = \frac{0 \cdot x^2 - 1 \cdot 2x}{x^4} = -\frac{2}{x^3}$~~
 ~~$\frac{d}{dx} \frac{1}{x^2} = \frac{0 \cdot x^2 - 1 \cdot 2x}{x^4} = -\frac{2}{x^3}$~~
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 So, $\frac{d}{dx} \frac{1}{x^2} = -\frac{2}{x^3}$
 after 10 minutes
 22. If $f(x) = x^2 + 3x + 2$ and $g(x) = x^2 + 2x + 1$, find $(f+g)(x)$.
 $(f+g)(x) = (x^2 + 3x + 2) + (x^2 + 2x + 1)$
 $(f+g)(x) = 2x^2 + 5x + 3$
 23. If $f(x) = x^2 + 3x + 2$ and $g(x) = x^2 + 2x + 1$, find $(f-g)(x)$.
 $(f-g)(x) = (x^2 + 3x + 2) - (x^2 + 2x + 1)$
 $(f-g)(x) = x^2 + 3x + 2 - x^2 - 2x - 1$
 $(f-g)(x) = x + 1$
 24. If $f(x) = x^2 + 3x + 2$ and $g(x) = x^2 + 2x + 1$, find $(fg)(x)$.
 $(fg)(x) = (x^2 + 3x + 2)(x^2 + 2x + 1)$
 $(fg)(x) = x^4 + 2x^3 + x^2 + 3x^3 + 6x^2 + 3x + 2x^2 + 4x + 2$
 $(fg)(x) = x^4 + 5x^3 + 9x^2 + 7x + 2$
 25. If $f(x) = x^2 + 3x + 2$ and $g(x) = x^2 + 2x + 1$, find $(\frac{f}{g})(x)$.
 $(\frac{f}{g})(x) = \frac{x^2 + 3x + 2}{x^2 + 2x + 1}$
 $(\frac{f}{g})(x) = \frac{(x+1)(x+2)}{(x+1)^2} = \frac{x+2}{x+1}$
 26. If $f(x) = x^2 + 3x + 2$ and $g(x) = x^2 + 2x + 1$, find $(\frac{g}{f})(x)$.
 $(\frac{g}{f})(x) = \frac{x^2 + 2x + 1}{x^2 + 3x + 2}$
 $(\frac{g}{f})(x) = \frac{(x+1)^2}{(x+1)(x+2)} = \frac{x+1}{x+2}$
 27. If $f(x) = x^2 + 3x + 2$ and $g(x) = x^2 + 2x + 1$, find $(f \circ g)(x)$.
 $(f \circ g)(x) = f(g(x)) = f(x^2 + 2x + 1)$
 $(f \circ g)(x) = (x^2 + 2x + 1)^2 + 3(x^2 + 2x + 1) + 2$
 $(f \circ g)(x) = x^4 + 4x^3 + 6x^2 + 4x + 1 + 3x^2 + 6x + 3 + 2$
 $(f \circ g)(x) = x^4 + 4x^3 + 9x^2 + 10x + 6$
 28. If $f(x) = x^2 + 3x + 2$ and $g(x) = x^2 + 2x + 1$, find $(g \circ f)(x)$.
 $(g \circ f)(x) = g(f(x)) = g(x^2 + 3x + 2)$
 $(g \circ f)(x) = (x^2 + 3x + 2)^2 + 2(x^2 + 3x + 2) + 1$
 $(g \circ f)(x) = x^4 + 6x^3 + 13x^2 + 12x + 5 + 2x^2 + 6x + 4 + 1$
 $(g \circ f)(x) = x^4 + 6x^3 + 15x^2 + 18x + 10$



Q. 1. A triangle ABC is formed by the vertices A(1, 2), B(4, 6) and C(7, 2). Find the area of the triangle.

Sol: The vertices of the triangle are A(1, 2), B(4, 6) and C(7, 2).
 The area of the triangle is given by:

$$\text{Area} = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

$$= \frac{1}{2} |1(6 - 2) + 4(2 - 2) + 7(2 - 6)|$$

$$= \frac{1}{2} |1(4) + 4(0) + 7(-4)|$$

$$= \frac{1}{2} |4 + 0 - 28|$$

$$= \frac{1}{2} |-24|$$

$$= \frac{1}{2} \times 24$$

$$= 12$$
 The area of the triangle is 12 square units.

Q. 2. A triangle ABC is formed by the vertices A(1, 2), B(4, 6) and C(7, 2). Find the perimeter of the triangle.

Sol: The vertices of the triangle are A(1, 2), B(4, 6) and C(7, 2).
 The perimeter of the triangle is given by:

$$\text{Perimeter} = AB + BC + CA$$

$$= \sqrt{(4-1)^2 + (6-2)^2} + \sqrt{(7-4)^2 + (2-6)^2} + \sqrt{(7-1)^2 + (2-2)^2}$$

$$= \sqrt{3^2 + 4^2} + \sqrt{3^2 + 16} + \sqrt{6^2 + 0}$$

$$= \sqrt{9 + 16} + \sqrt{9 + 16} + \sqrt{36}$$

$$= \sqrt{25} + \sqrt{25} + 6$$

$$= 5 + 5 + 6$$

$$= 16$$
 The perimeter of the triangle is 16 units.

Q. 3. A triangle ABC is formed by the vertices A(1, 2), B(4, 6) and C(7, 2). Find the orthocenter of the triangle.

Sol: The vertices of the triangle are A(1, 2), B(4, 6) and C(7, 2).
 The orthocenter of the triangle is the point where the altitudes intersect.
 The altitude from A to BC is perpendicular to BC. The slope of BC is $\frac{2-6}{7-4} = -\frac{4}{3}$. The slope of the altitude from A is $\frac{3}{4}$. The equation of the altitude from A is $y - 2 = \frac{3}{4}(x - 1)$.
 The altitude from B to AC is perpendicular to AC. The slope of AC is $\frac{2-2}{7-1} = 0$. The slope of the altitude from B is undefined. The equation of the altitude from B is $x = 4$.
 The orthocenter is the intersection of the altitudes $y - 2 = \frac{3}{4}(x - 1)$ and $x = 4$.
 Substituting $x = 4$ into the equation of the altitude from A:
 $y - 2 = \frac{3}{4}(4 - 1)$
 $y - 2 = \frac{3}{4} \times 3$
 $y - 2 = \frac{9}{4}$
 $y = \frac{9}{4} + 2$
 $y = \frac{9}{4} + \frac{8}{4}$
 $y = \frac{17}{4}$
 The orthocenter is $(4, \frac{17}{4})$.

Q. 1. A triangle ABC is formed by the vertices A(1, 2), B(4, 3) and C(3, 6). Find the area of the triangle.

Sol: The vertices of the triangle are A(1, 2), B(4, 3) and C(3, 6).
 The area of the triangle is given by:

$$\text{Area} = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

$$= \frac{1}{2} |1(3 - 6) + 4(6 - 2) + 3(2 - 3)|$$

$$= \frac{1}{2} |-3 + 20 - 3|$$

$$= \frac{1}{2} |14|$$

$$= 7 \text{ sq. units}$$

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Q. 4. A triangle ABC is formed by the vertices A(1, 2), B(4, 3) and C(3, 6). Find the area of the triangle.

Sol: The vertices of the triangle are A(1, 2), B(4, 3) and C(3, 6).
 The area of the triangle is given by:

$$\text{Area} = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

$$= \frac{1}{2} |1(3 - 6) + 4(6 - 2) + 3(2 - 3)|$$

$$= \frac{1}{2} |-3 + 20 - 3|$$

$$= \frac{1}{2} |14|$$

$$= 7 \text{ sq. units}$$

(iv) Mid point of PR

$$\left(\frac{2+12}{2}, \frac{4+16}{2} \right)$$

$$\left(\frac{2+12}{2}, \frac{4+16}{2} \right)$$

$$\left(\frac{10}{2}, \frac{20}{2} \right)$$

$$(5, 10)$$

(v) let $A(x, y)$ be point on x-axis
which is equidistant from P & Q
so $PA = QA$
 $PA^2 = QA^2$

$$(2-x)^2 + (5-0)^2 = (4-x)^2 + \left(\frac{13}{3}-0\right)^2$$

$$4 + 2x^2 - 4x + 25 = 16 + x^2 - 8x + \frac{169}{9}$$

$$4x = 16 + \frac{169}{9} - 4 - 25$$

$$x = \frac{13}{9}$$

Hence the point is $\left(\frac{13}{9}, 0 \right)$

quadratic poly. $p(x) = 2x^2 - 3x + 7$

here $a = 2$

$b = -3$

$c = 7$

$$x + B = -\frac{b}{a} = -\frac{-3}{2} = \frac{3}{2}$$

$$6x + B = \frac{7}{2}$$

$$\frac{1}{2x-3} + \frac{1}{2B-3}$$

$$\frac{(2x-3) + (2B-3)}{(2x-3)(2B-3)}$$

$$\frac{2B-3 + 2x-3}{4xB - 6x - 6B + 9}$$

$$\frac{2B + 2x - 6}{4xB - 6(x+B) + 9}$$

$$\frac{2(x+B) - 6}{4xB - 6(x+B) + 9} = \frac{2x+B-3}{4xB-6x-6B+9}$$

$$\frac{2x \cdot \frac{3}{2} - 6}{4x \cdot \frac{7}{2} - 6 \cdot \frac{3}{2} + 9} = \frac{3-6}{14-9+3} = \frac{-3}{14}$$

$$\text{Ans. } \frac{1}{2x-3} + \frac{1}{2B-3} = \frac{-3}{14}$$